

Chapter 16

Adoption of Computer– Based Formative Assessment in a High School Mathematics Classroom

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EXECUTIVE SUMMARY

This case follows a high school mathematics teacher who is new to the classroom and is looking to adopt computer-based formative assessment as a part of his curriculum. Working within the confines of the school environment, this requires navigating a shrinking budget, colleagues that do not share his value of technology, restricted time, student issues, and limited resources. He must examine all aspects of the available computer-based formative assessment systems and weigh the pros and cons to insure the best academic outcomes for his students.

ORGANIZATION BACKGROUND

Wilderness Central School District is a public district serving approximately 3,500 students in grades Pre K-12. The district was created in 1948 when five smaller districts were consolidated in the interest of centralizing and sharing resources. The district is located in an area classified as “rural fringe” by the U.S. Department of Education Institute of Education Sciences in the northeast region of the United States near the

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state capital. This area has seen a steady population increase as new industries based in technology have moved into the region bringing jobs and business opportunities. Employees of these new companies do not wish to live within city limits, preferring more suburban neighborhoods as well as quality schools. The school district of the state capital has been facing budget cuts and decreased state aid for the past five years, leading to limitations in resources. Like many inner-city schools, the capital city district has difficulty recruiting and retaining quality teachers. Newcomers to the area who possess the means often choose to live outside the reaches of the city school district. A majority of the faculty and staff working at Wilderness CSD live in the district while some, living in adjoining districts, exercise an option offered by Wilderness to bring their children to the district tuition-free. Employees of the district have a great deal of pride in their schools and their community.

As a public school district, Wilderness is open to all students living within the boundaries of the district. The demographic profile of the district is quite homogeneous and the majority of the residents are white. Due to the association with the local industries, most households fall at about the median income level for the state. As such the state has categorized the district as an average needs district using a ratio of available resources to resources necessary to provide for the needs of all students.

The school district shares a similar makeup to the surrounding community. Residents of the district pay school taxes which, when combined with aid from the state, supports the school's budget. Like the capital city district, Wilderness has seen cuts in state aid in recent years and has been forced to rely on reserve funds to avoid a large school tax increase. However, sustained cuts have resulted in the depletion of reserve funds and boosted the tax burden of the district residents. Residents are evenly divided on the issue and school budget votes have been decided by relatively few votes for the past three years.

The district's high school, Wilderness Senior High School, serves approximately 1,200 students in grades 9-12. The school underwent restorations around ten years ago to address structural issues. As the state funds were much more available at that time, Wilderness High received a grant to rebuild their football field with a running track, add a wing with ten new classrooms, and build an updated rehearsal studio for the school band. At that time, several parents and community members petitioned the school board to invest some of the funds received into technology for the high school. However, as the school building required construction to fix the structural issues, the board decided that the funds from the state would be used to expand the scope of the building project.

Since passing on the opportunity to use state funds to bolster technology in the school, Wilderness High has attempted to keep up with advances in educational technologies. Unfortunately the school has fallen behind due to a lack of drive to include technology in the curriculum and unavailability of funding. The school

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librarian is responsible for managing what technology the school possesses, which consists of two computer labs accommodating 30 students each and ten LCD projectors that can be checked out by teachers with advance notice. These resources must be shared between approximately 110 faculty members. Additionally, all teachers have either a laptop or desktop computer that is used for attendance and student data management. Approximately one-third of classrooms, predominantly science labs and English classrooms, have at least one computer available for student use.

Bill Rogers is a recent graduate of a teacher education master's degree program at a small public university in the northeastern United States. Prior to this, he earned a bachelor's degree in mathematics at the same university. As a part of his teacher education program, Bill took courses in assessment and educational technology. In his assessment course he learned about formative assessment which describes a system of assessment for learning as opposed to assessment of learning (Stiggins, Artur, Chappuis, & Chappuis, 2006). Bill knows that students are routinely subjected to high-stakes standardized tests to summatively assess their knowledge. He hopes that he will have the opportunity to implement formative assessments in his classroom that help students progress in their learning instead of just evaluate their status.

For most of his life Bill has been interested in technology. He enjoys working with computers and is fluent with many popular software titles as well as basic hardware issues. Taking a course in educational technology, Bill was excited to learn about all of the ways that technology can enrich the curriculum in a classroom. As a new hire tasked with teaching algebra and geometry at Wilderness High, he hopes to use what he has learned to incorporate computer-based formative assessment into his classroom to inform his instruction and help in monitoring student progress toward academic achievements.

SETTING THE STAGE

Formative Classroom Assessment in Mathematics

In his synthesis of research on achievement, John Hattie (2009) concluded that “the biggest effects on student learning occur when teachers become learners of their own teaching and when students become their own teachers” (p. 22). This suggests an advocacy for two distinct strategies to improve learning. A teacher reflecting on their own teaching allows for a view of the classroom from the student perspective leading to tailored instruction that best supports student achievement. At the same time, students taking ownership of their learning promotes the attributes of self-regulation which have shown time and time again to produce higher achievement by students (Zimmerman & Schunk, 2001). This dual-strategy approach is

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supported quite well in the scholarship of formative classroom assessment. Indeed, Cizek (2010) called the latest conception of formative assessment a “collaborative process engaged in by educators and students for the purpose of understanding the students’ learning and conceptual organization, identification of strengths, diagnosis of weakness, areas for improvement, and as a source of information that teachers can use in instructional planning and students can use in deepening their understandings and improving their achievement” (pp. 6-7).

Hattie’s (2009) claim that half of the equation leading to maximum student achievement comes from teachers examining their teaching is well-supported in his synthesis of meta-analyses. In fact, when ranked by effect size, six of the top ten influences on student achievement were contributions from the teacher. Formative assessment was the highest-ranked of these practices with an average effect size of 0.90. Black and Wiliam’s (1998) review of classroom practices revealed that modern classroom assessments were “beset with problems and shortcomings” (p. 141). Specifically, the authors found that current practices did not provide information to students to help them improve their work, resulting in superficial learning and valued quantity over quality. The assessments teachers were using in their classrooms left students at the lowest levels of achievement believing they were not able to learn or work at higher levels. What came of these findings was a shift in focus towards formative assessments. Formative assessment is best thought of as assessment *for* learning and is contrasted with more traditional summative assessments which may be thought of as assessment *of* learning (Stiggins, Artur, Chappuis, & Chappuis, 2006). Formative assessments occur here and now in the classroom and provide feedback to students about the quality of their work and their levels of understanding. At the same time, teachers receive feedback about specific student learning needs. Both groups have the opportunity to be given feedback about their strengths and weaknesses of their work and a subsequent chance to revise and improve the product (Cizek, 2010). Wiliam (2007) estimated that formative assessment interventions can produce an average six to nine months of learning gained per year at a cost that is approximately one-tenth of that associated with reducing class size by 30%, which results in only three months gained on average.

Andrade (2010) claimed that the definitive source of formative feedback is the student in the classroom as students have “constant and instant access to their own thoughts, actions, and works” (p. 90). She describes student self-assessment as a process where students “reflect on the quality of their work, judge the degree to which it reflects explicitly stated goals or criteria, and revise accordingly” (p. 92). This research is anchored in self-regulated learning and feedback in learning. A great deal of scholarship has suggested that self-regulation is closely related to academic achievement, specifically positing that students who set goals, develop plans to meet them, and monitor their progress along the way tend to learn more

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and succeed more readily in scholastic endeavors than students who do not engage in these practices. Students with less developed self-regulation abilities are forced to place a higher dependence on other sources such as the teacher, fellow students, or task-specific qualities in order to receive necessary feedback (Pintrich, 2000; Zimmerman & Schunk, 2004).

Supplying students with feedback to be used in improving the products of their learning, as well as their understanding of the subject matter, is a main goal of formative assessment. Feedback has the power to close the gap between where a student is currently and where they aspire to be (Sadler, 1989). This finding caused Hattie (2009) to conclude that feedback can be “very powerful in enhancing learning” (p. 178). Wiliam and Thompson (2007) note that the most effective feedback provides information specific to the task and about the goals of student learning, where the student is in relation to their particular goals, and what can be done to close this gap. Students tend to get very little feedback on their classwork (Black & Wiliam, 1998) despite findings that demonstrate the ability of correctly delivered feedback to enhance learning and improve academic achievement (Bangert-Drowns, Kulik, Kulik, & Morgan, 1991; Shute, 2008). Hattie and Timperly (2007) reported that a lack of feedback can often be attributed to the fact that many teachers do not have the time necessary to quickly and frequently react and offer feedback for work from a full class of students. At the same time students and teachers both assume that the responsibility for feedback rests solely on the teacher. Despite any feelings from stakeholders that teachers should be the primary, or even sole, source of feedback for students in the classroom, students themselves can provide useful and effective feedback (Andrade, 2010; Andrade, Du, & Wang, 2008; Andrade, Du, & Mycek, 2010).

Formative assessment in mathematics is currently an especially relative topic as students are now expected to demonstrate mathematical competency that moves beyond proficiency in computational exercises to extended responses to problems (National Council of Teachers of Mathematics, 2000). Superficial abilities are no longer accepted by stakeholders and a conceptual understanding of math is required to succeed in school and beyond. Still, state, national and international reports indicate students often do not demonstrate mathematical reasoning at the necessary level (National Center for Educational Statistics, 2008; New York State Education Department, 2008). Formative assessment in mathematics classrooms has not received a great deal of attention in either theoretical or practical literature. Yet, an examination of what little research exists, along with parallel investigations in other contents areas, demonstrates that formative assessment in mathematics can help students improve their mathematical reasoning skills which, in turn, will lead to higher achievement in math. Several international studies showed evidence for the positive impact of formative assessment on student performance in mathematics. Fontana and Fernandes (1994) worked with Portuguese middle and high school

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mathematics students using an intervention consisting of multiple formative strategies. The investigators found improvements in students' academic performance using this treatment. Black, Harrison, Lee, Marshall, and Wiliam (2004) studied formative assessment practices in middle and high school math and science classes in England and demonstrated a strong relationship between these formative assessment practices and scholastic achievement.

Black and Wiliam (1998) called for policy changes that allowed students to “take active responsibility for their own learning” (p. 146). At the same time, teachers need to take responsibility for what they teach. Both parties need to regulate and reflect on what they are doing. Research from educational psychology supports that when both components are in action, the largest effects on student achievement will occur (Hattie, 2009). Hattie adds that “learning is a very personal journey for the teacher and the student, although there are remarkable commonalities in the journey for both” (p. 23). These commonalities are found in metacognitive and classroom strategies that can be employed to take ownership of teaching and learning. Effective teachers are able to not only utilize these strategies, but model them and teach them to their students so that the next generation of learners is prepared to take responsibility for their learning.

Computer-Based Formative Assessment Technology

Maier and Warren (2000) suggested that assessment strategies have fallen behind pedagogical innovations, but that interest in assessing students in ways that support diverse learning outcomes is on the rise. Over a decade ago, Stephens, Bull and Wade (1998) warned that technology should not drive the assessment process. Indeed, technology innovations are most effectively used to support assessment strategies and contribute to their authenticity. Pellegrino (2006) predicted that as technology continues to influence education in important ways, assessments would develop stronger links with instruction moving towards full integration. Computer-based or computer-assisted assessment is a tool that can help strengthen these links, “particularly where [it] is combined with useful feedback” (Dalziel, 2001, p. 1). As discussed above, feedback is a crucial part of formative assessment and computer-based assessment allows for instantaneous feedback on student work. Specifically, formative assessment technologies allow for the measurement of “how well students can *produce*, rather than choose, right answers, giving evaluative and instructive feedback immediately, automatically choosing the best next assignments and assessments, and providing richer information to guide teacher interventions” (Landauer, Lochbaum, & Dooley, 2009, p. 45). The benefits of computer-based assessment also include less distance between assessments and learning environments (Brown et al., 1999).

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While avoiding delving into specific software programs, Dalziel described five pedagogical models which illustrate the potential uses computer-based assessment for formative purposes. They include:

1. The use of pre-tests, taken prior to learning to assist in determining levels of existing understanding and appropriate courses for study;
2. The use of self-testing “objects” within learning (where a “mini-test” is embedded within a given page in a larger course);
3. The use of “end of section” tests, which are generally larger than mini-tests, and may focus on a greater breadth of content as part of an attempt to consolidate learning across a given content area;
4. Different types of feedback, such as multi-layered feedback, option specific feedback, the ability to try again, etc;
5. The use of final quizzes at the end of learning, particularly where these draw from a bank of questions, and hence encourage more than one attempt at a final test (Dalziel, 2001, p. 4).

For over a decade, various assessment companies have produced software programs that have students complete test items which may be used to support more traditional assessments. The software provides real-time feedback about responses to items to both the students and to the teacher (Penuel & Yarnall, 2005). There is empirical evidence that students’ academic achievement increases when assessment software is used for formative purposes (e.g., Charman & Elmes, 1998; Sly & Rennie, 1999). Additional research has shown that Web-based formative assessment can help increase student academic achievement in an e-Learning environment (e.g., Buchanan, 2000).

Pointing out that the sheer number of students to be attended to can make formative assessment a weighty task in a traditional classroom, Bransford, Brown, and Cocking (2000) suggested that an e-Learning environment is rife with opportunities for formative assessment and student self-assessment which is an important component (e.g., Andrade, 2010). Wang (2008) reviewed the structure of several formative assessment software and web-based programs and conclude that the inclusion of four distinct strategies is necessary for effective integration of computer-based formative assessment: (1) multiple administrations of formative tests; (2) rapid feedback that does not automatically give students the correct answer; (3) access to score history for students and teachers; and (4) the ability to ask questions through email or other means. Vogel, Greenwood-Ericksen, Cannon-Bower, and Bowers (2006) also weigh in with the finding that the opportunity to interact with a computer may motivate students’ desire to learn, so a formative assessment that models the act of playing a game is likely to engage students above other designs.

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There are some challenges associated with incorporating computer-assisted assessment into the classroom. Black and colleagues (2003) reported that some of the teachers they work with found the software use to be too time consuming due to the volume of student progress data collected. McKenna and Bull (2000) pointed out that computer-based assessment integration requires the devotion of time and fiscal resources to be effective. Still, Collins and Halverson (2009) insist that today's students who have been raised with technology in every part of their lives will not sit idly by as schools refuse to embrace new innovations. Learning is moving to a digital realm and if stakeholders hope to discover the extent of that learning, assessment must move to that same playing field.

Computer-Based Formative Assessment in Mathematics

Content-specific formative assessment technologies are most readily available in language arts, math, and science disciplines. According to Landauer, Lochbaum, and Dooley (2009), these subjects lend themselves to a computer-based assessment because they “involve dynamic activities where errors and successes happen almost continuously and probably should, at least sometimes, be assessed very quickly and frequently for maximum benefit” (p. 52). Based on previous work with his colleagues (i.e., Wang et al., 2004) creating the Web-based Assessment and Test Analysis System (WATA), Wang (2011) developed a Web-based dynamic assessment system for junior high school mathematics students requiring remediation. The system was based on the theory of dynamic assessment which uses Vygotsky's (1978) “Zone of Proximal Development” and assumes that students can achieve higher levels of cognition with the support of those around them, be they other students or the teacher. Ashton and colleagues (2006) proposed a computer-based assessment system that awarded partial credit to students if they were able to respond correctly to items that were created by breaking up the original question into smaller pieces.

Other than the systems created by researchers and non-profit organizations, are those computer-based formative assessment programs created by private companies. The Children's Progress Academic Assessment (2010) provides formative assessment of students' skills in math as well as language arts. The system is adaptive and generates both evaluation reports and suggestions for the teacher regarding instructional decisions. Brainchild's (2009) *Achiever!* is an online formative assessment system that pretests students and offers instruction and study plans that are tailored to specific student needs. Teachers have the option of creating assignments or using those automatically generated by the system. In a similar vein, HeartBeeps (2011) software provides formative and summative assessment that is tied directly to the curriculum. Teachers are able to view current proficiency levels for each individual student and remediate as necessary. Computer-based assessments such as these pro-

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vide students with feedback and teachers with information that they can use to make instructional decisions, both of which are primary goals of formative assessment.

The availability and potential benefits of computer-based formative assessment software are well supported by both theoretical and practical literature. The challenges that a prospective user may face occur during the implementation phase where the software is obtained, prepared for use, and utilized by the students and teacher. A great deal of educational research has demonstrated that multiple variables exist and interact within the classroom. This can make introducing a new intervention designed in a laboratory or environment other than the specific classroom where it will be used especially challenging (Confrey, 2006).

Case Description

Having been hired by Wilderness Senior High School in June, Bill spent a good deal of time over the summer evaluating various computer-based formative assessment programs, both Web- and software-based. He weighed the pros and cons of each system, specifically looking at ease of use, ability to inform instruction, type of feedback given to students, and resources necessary for implementation. Ties to the state curriculum were also an important issue that Bill considered while deliberating. After careful consideration Bill chose a software-based formative assessment system for use in his classroom. He particularly liked that the software had a pretest component that judged students' current levels of proficiency in terms of grade-level state standards. Bill hoped to have all of his students take this pretest within the first two weeks of the school year so that he would have an understanding of each individual student's academic standing in mathematics.

As a new employee of the district, Bill attended a two-day orientation session directly before the start of the school year. Here he was introduced to the administrative aspects of his job such as using the student record software to take attendance and keep student grades. He also attended a session on the day-to-day operations of Wilderness Senior High School which included information on the daily schedule, a tour of the building, and an overview of the contact people for various resources. During this time, Bill noticed the lack of computers in classrooms. When the tour stopped at the room that was to be his, Bill was disappointed to note that there were no computers for student use.

Following the district orientation, Bill had an opportunity to meet with members of the math department at Wilderness High. The ten member department consisted of nine veteran teachers and another new graduate who was hired at the same time as Bill. During this meeting, Bill brought up the lack of computers and his plan for pretesting students using formative assessment software. Remarking that there would likely not be time for that, the department head used Bill's inquiry as a segue

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to hand out and discuss curriculum coverage maps for the year. Bill was dismayed to see that topics to be taught were sorted by semester and even roughly assigned certain date ranges. He was scheduled to begin teaching new content on the first day of school with no time to assess students' current understandings.

As a new teacher, Bill had been assigned a mentor. This was a veteran math teacher who was to help him acclimate to his new career and assist him in navigating the new surroundings. Bill took his concerns about the lack of pretesting time to his mentor. She was sympathetic to his derailed planning, but unsure of his motives. She questioned why he would need to assess students before teaching anything unless he was attempting to scrutinize the students' previous math teacher. Assuring her that he was only trying to gauge the current proficiency levels of his students to inform his teaching, Bill explained the principles of formative assessment to his mentor. While she knew about formative classroom assessment from professional development and graduate classes, Bill's mentor had never been clear on what it would look like in a high school math classroom. She was still skeptical, but, after seeing Bill's enthusiasm, allowed him to take the student computer that was in her computer because "we never used it anyway." However, as a caveat, she cautioned Bill that he may not want to spread his plan around as other faculty may not understand or appreciate his intentions and might view them as commentary on their teaching abilities.

Bill took the computer given to him and set it up in his classroom. The computer was several years old and had clearly not been turned on in the past school year. Bill wiped the dust off of the case and peripherals and turned it on. After running some diagnostic tests, Bill determined that the computer met all of the hardware requirements to run his chosen software. He was concerned about the speed of the computer, but realized that he was lucky to have gotten it. Now, with access to a computer for student use, Bill was free to obtain the software. He approached his department head to seek funding to purchase a copy and was told that the department had very little in the way of discretionary funds and they had already been used to purchase classroom sets of graphing calculators required to the state-level standardized assessments. Like his mentor, the department head questioned the need for formative assessment software and reminded Bill that the curriculum maps must be followed so that students can prepare for state tests. His demeanor was friendly, but he suggested that Bill may want to focus on the material to be taught instead of bringing in new pedagogical tools.

Not wanting to cause any unrest with his colleagues before the school year even began, Bill asked the department head if he might pursue funding with the principal. The department head gave his blessing, but advised "don't get your hopes up." Bringing his request to the principal, Bill received the same answer – the school's

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resources are depleted due to mandated spending and budget cuts. Having studied assessment more deeply than many teachers, the principal was supportive of Bill's idea and, while he could not offer any funding sources, Bill might consider contacting the software developers, explaining the budgetary constraints, and asking if there was anything the company could do.

Frustrated, but still believing that formative classroom assessment would benefit the students he had yet to meet, Bill contacted the software corporation. He was directed to their customer service division where he was able to explain his situation to a company representative. Since Bill is only requesting a single license, the representative offered him the software for free if he will provide feedback to the developers and share his user experience with them. The representative explains that the company is trying to reach more institutions with their products and Bill's experience, combined with his word-of-mouth recommendation will help them with that goal. Bill received the setup files for the software via email later that day.

Bill installed the software on the computer and was fortunate to avoid any hardware compatibility incidents. He used his class list to set up individual accounts for all of his students that they could access to monitor their progress. In the final days before the start of the school year, Bill planned his lessons for the first week of classes. He developed a schedule that allowed students to take turns at the computer and complete the pretest. Once he felt prepared, he awaited the first day of school, feeling more confident than ever that using computer-based formative assessment would give him an edge in making instructional decisions and would help students self-regulate their learning leading to improved academic achievement.

Current Challenges/Problems Facing the Organization

During the first week of school, Bill is attempting to follow his plans, including his schedule for having students take the pretest on the single computer in his classroom. Student absences have made this very difficult and Bill has had to rearrange the schedule daily. Another obstacle Bill faces is that the pretest takes longer to complete than he originally planned. Students take time to understand the task and the user interface of the formative assessment program. This further erodes the predetermined schedule. While the students have been fairly compliant so far, Bill has spoken to several students who complained about taking the pretest, seeing it as a "test for no reason." He has tried to explain the basic tenants of formative assessment and self-regulated learning, but is concerned that some students may not be putting forth their full effort when taking the pretest. If this is the case, the levels of understanding for each student will not be accurate and Bill cannot use them to inform his future instruction.

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Beyond the student issues, Bill is fielding questions from a variety of faculty members regarding his use of technology. The questions seem to be in two distinct veins. One group is genuinely curious about his use of computer-based formative assessment and may want to try a similar undertaking if the results are positive. The other group feels that Bill is “rocking the boat” by introducing a new dimension of instruction. They are concerned about the implication that their current methods are ineffective. This is furthered by the building principal’s vocal support of Bill’s work. Faculty members have always been professional, but Bill feels some silent opposition, even within his own department.

An additional issue has been allocation of resources. It is crucial to their development of self-regulation skills that they are able to monitor their progress and see where any gaps exist. Bill has asked that additional software licenses be purchased and installed on the computers in one of the school’s two labs. He has also tried to reserve a computer lab once per week so that students can view the results of their pretest and subsequent formative assessments to keep updated on their progress towards their academic goals. Once again, he is told that no money exists to purchase additional software. A customer service representative at the software manufacturer explains that it would be too expensive to give out 30 free licenses to one school. Furthermore, the school librarian has informed Bill that his request for the lab cannot be accommodated. She reminded him that there are only two labs to be shared and Bill cannot monopolize them. Requests for additional computers for his classroom go unanswered.

Bill knows that if he can get an accurate picture of each student’s proficiency in mathematical concepts, he can tailor instruction to meet their needs. He is confident that the formative assessment software he has been using can judge these levels and can help students take charge of their learning by monitoring their progress towards an academic goal. However, until he can have every student give their best effort on the pretest and have access to their individual account with assessment data, it will be very difficult to make instructional decisions beyond what has been established by the department’s curriculum map. Unfortunately, this map does not offer any strategies for filling the gaps that exist in students’ understanding of content that was covered in previous years. Only the results of the pretest can provide that information.

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All locations and characters in this case are fictional.

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